FLAT PULLBACK OF THE DIVISOR OF A MEROMORPHIC SECTION OF AN INVERTIBLE SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Additivity of the divisor of a meromorphic function. Divisors, label divisors-lemma-c1-additive, invoked at proof line 4514. Reduces the statement for a general meromorphic section written as a quotient of two global sections to the two statements for those sections.
- The cycle identity for the zero scheme of a section and its restriction to the components. Same chapter chow.tex, the equation labelled equation-equal-as-cycles, invoked at proof line 4517 and established by S14-algebra-prepare-geometric-cap. Identifies the class of the zero scheme of the pulled back section with the sum over the components of the divisors of the restricted sections.
- The cycle class of a divisor and flat pullback of cycle classes of coherent sheaves. Same chapter chow.tex, labels lemma-compute-c1 (proof line 4528) and lemma-pullback-coherent (proof line 4528, claimed as S14-algebra-pullback-coherent). Rewrites the flat pullback of the divisor as the flat pullback of a cycle class and then as the class of the inverse image, which coincides with the zero scheme of the pulled back section.

Source reference. Original lemma 02SL: sources/stacks-chow/tags.txt, source lines 3664–3664. Navigation pointer for \ref{divisors-lemma-c1-additive} used inside the counted span (line 4514); the argument that consumes it is retained in the span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original equation 02SR: sources/stacks-chow/pinned-original-source.tex, source lines 4375–4375. Navigation pointer for \ref{equation-equal-as-cycles} used inside the counted span (line 4517); the argument that consumes it is retained in the span.

Source reference. Original lemma 02SK: sources/stacks-chow/pinned-original-source.tex, source lines 4198–4198. Navigation pointer for \ref{lemma-compute-c1} used inside the counted span (line 4528); the argument that consumes it is retained in the span.

Source reference. Original lemma 02RE: sources/stacks-chow/pinned-original-source.tex, source lines 2331–2331. Navigation pointer for \ref{lemma-pullback-coherent} used inside the counted span (line 4528); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 4485); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X, Y be locally of finite type over S. Let $f : X \to Y$ be a flat morphism of relative dimension r. Let $\mathcal{L}$ be an invertible sheaf on Y. Assume Y is integral and $n = \dim_\delta(Y)$. Let s be a nonzero meromorphic section of $\mathcal{L}$. Then we have*

$$f^* div_{\mathcal{L}}(s) = \sum n_i div_{f^*\mathcal{L}|_{X_i}}(s_i)$$

in $Z_{n+r-1}(X)$. Here the sum is over the irreducible components $X_i \subset X$ of δ-dimension $n + r$, the section $s_i = f|^*_{X_i}(s)$ is the pullback of s, and $n_i = m_{X_i,X}$ is the multiplicity of X_i in X.

Proof. To prove this equality of cycles, we may work locally on Y. Hence we may assume Y is affine and $s = p/q$ for some nonzero sections $p \in \Gamma(Y, \mathcal{L})$ and $q \in \Gamma(Y, \mathcal{O})$. If we can show both

$$f^*\mathrm{div}_{\mathcal{L}}(p) = \sum n_i \mathrm{div}_{f^*\mathcal{L}|_{X_i}}(p_i) \quad \text{and} \quad f^*\mathrm{div}_{\mathcal{O}}(q) = \sum n_i \mathrm{div}_{\mathcal{O}_{X_i}}(q_i)$$

(with obvious notations) then we win by the additivity, see Divisors, Lemma 02SL. Thus we may assume that $s \in \Gamma(Y, \mathcal{L})$. In this case we may apply the equality (02SR) to see that

$$[Z(f^*(s))]_{k+r-1} = \sum n_i \mathrm{div}_{f^*\mathcal{L}|_{X_i}}(s_i)$$

where $f^*(s) \in f^*\mathcal{L}$ denotes the pullback of s to X. On the other hand we have

$$f^*\mathrm{div}_{\mathcal{L}}(s) = f^*[Z(s)]_{k-1} = [f^{-1}(Z(s))]_{k+r-1},$$

by Lemmas 02SK and 02RE. Since $Z(f^*(s)) = f^{-1}(Z(s))$ we win. □